

Michael Crawshaw

Research Fellow

Flatiron Institute, Center for Computational Mathematics

Research Area: Optimization for machine learning

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🌐 Personal Website

🔗 Google Scholar

EDUCATION

George Mason University

Ph.D. in Computer Science

2019 - 2026

M.S. in Computer Science

2019 - 2022

Advisor: Mingrui Liu

The Ohio State University

B.S. in Mathematics and Computer Science, Honors

2015 - 2019

PUBLICATIONS

FlashBoB: I/O-Efficient Exact Backward-over-Backward for Softmax Attention

Anthony Givans, **Michael Crawshaw**, Mingrui Liu

Conference on Neural Information Processing Systems, 2026.

Muon Does Not Converge on Convex Lipschitz Functions

Tetiana Parshakova, Ahmed Khaled, **Michael Crawshaw**, Guillaume Garrigos, Robert M. Gower

Conference on Neurips Information Processing Systems, 2026.

An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants

Michael Crawshaw, Chirag Modi, Mingrui Liu, Robert M. Gower

International Conference on Machine Learning, 2026.

Non-Euclidean Gradient Descent Operates at the Edge of Stability

Rustem Islamov, **Michael Crawshaw**, Jeremy Cohen, Robert M. Gower

International Conference on Machine Learning, 2026 (oral).

Tight Bounds for Logistic Regression with Large Stepsize Gradient Descent in Low Dimension

Michael Crawshaw, Mingrui Liu

Conference on Learning Theory, 2026.

Constant Stepsize Local GD for Logistic Regression: Acceleration by Instability

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Machine Learning, 2025.

Local Steps Speed up Local GD for Heterogeneous Distributed Logistic Regression

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Learning Representations, 2025.

Complexity Lower Bounds of Adaptive Gradient Algorithms for Non-convex Stochastic Optimization under Relaxed Smoothness

Michael Crawshaw, Mingrui Liu

International Conference on Learning Representations, 2025.

Federated Learning under Periodic Client Participation and Heterogeneous

Data: A New Communication-Efficient Algorithm and Analysis

Michael Crawshaw, Mingrui Liu

Conference on Neural Information Processing Systems, 2024.

Provable Benefits of Local Steps in Heterogeneous Federated Learning for Neural Networks: A Feature Learning Perspective

Yajie Bao, **Michael Crawshaw**, Mingrui Liu

International Conference on Machine Learning, 2024.

Federated Learning with Client Subsampling, Data Heterogeneity, and Unbounded Smoothness: A New Algorithm and Lower Bounds

Michael Crawshaw^{*}, Yajie Bao^{*}, Mingrui Liu (^{*} denotes equal contribution)

Conference on Neural Information Processing Systems, 2023.

EPISODE: Episodic Gradient Clipping with Periodic Resampled Corrections for Federated Learning with Heterogeneous Data

Michael Crawshaw, Yajie Bao, Mingrui Liu

International Conference on Learning Representations, 2023.

Robustness to Unbounded Smoothness of Generalized SignSGD

(Alphabetical order) **Michael Crawshaw**, Mingrui Liu, Francesco Orabona, Wei Zhang, Zhenxun Zhuang

Neural Information Processing Systems, 2022.

Fast Composite Optimization and Statistical Recovery in Federated Learning

Yajie Bao, Michael Crawshaw, Mingrui Liu

International Conference on Machine Learning, 2022.

EMPLOYMENT

Flatiron Institute

September 2026 - Present

Research Fellow, Center for Computational Mathematics

- Theoretical and empirical research in optimization for machine learning.
- Particular focus on non-Euclidean optimization and unstable optimization in deep learning.

American University

June 2026 - August 2026

Adjunct Lecturer, Department of Computer Science

- Taught a seven-week summer course titled CSC 124: Exploring AI Through Programming.
- With minimal prerequisites, students learned about AI from scientific, ethic, political, and economic perspectives.

Flatiron Institute

May 2025 - August 2025

Summer Predoctoral Researcher, Center for Computational Mathematics

- Developed novel non-Euclidean gradient descent methods to train deep neural networks, especially language models.
- Proposed methods match or outperform state-of-the-art with significantly improved robustness to hyperparameter tuning.

Olive

March 2018 - August 2019

Machine Learning Engineering Intern

- Developed computer vision functionality for desktop automation software with applications to healthcare operations.
- Trained deep neural networks for object detection with various techniques, including Faster R-CNN and DARTS.

AWARDS

Outstanding Dissertation Award, Department of Computer Science, George Mason University

2026

Silver Reviewer, ICML

2026

Top Reviewer, NeurIPS

2025

Top Reviewer, ICML

2025

Best Reviewer, AISTATS

2025

Top Reviewer, NeurIPS (top 8%)

2023

Institute for Digital Innovation Predoctoral Fellowship, George Mason University

2022

NSF XSEDE startup allocation, National Science Foundation

2020

Summer Ph.D. Research Initiation Award, George Mason University

2020

Outstanding Graduate Teaching Assistant, George Mason University

2020

Gordon Memorial Fund Scholarship, The Ohio State University

2017 - 2019

Honorable Mention, Raser-Bareis-Gordon Math Competition, The Ohio State University

2017

7th Place, FIRST Tech Challenge World Competition, FIRST

2015

ACADEMIC SERVICE

Reviewer, TMLR, AAAI, ICLR, ICML (awarded Silver Reviewer), NeurIPS, JMLR

2026

Reviewer, ICLR, AISTATS (awarded Best Reviewer), ICML (awarded Top Reviewer), NeurIPS (awarded Top Reviewer)

2025

Reviewer, AISTATS, ICML, NeurIPS

2024

Reviewer, NeurIPS (awarded Top Reviewer, top 8%)

2023

TEACHING

Adjunct Lecturer, American University

2026

CSC 124: Exploring AI Through Programming

Summer 2026

Graduate Teaching Assistant, George Mason University

2019-2022

CS 657: Mining Massive Datasets

Fall 2020, Fall 2021

CS 471: Operating Systems

Fall 2020, Fall 2021, Spring 2022

CS 583: Analysis of Algorithms

Spring 2021

CS 571: Operating Systems

Spring 2021

CS 330: Formal Methods and Models

Fall 2019, Spring 2020

Undergraduate Teaching Assistant, The Ohio State University

2017 - 2018

CSE 3321: Automata and Formal Languages

Summer 2017, Fall 2017, Spring 2018

Undergraduate Honors Math Mentor, The Ohio State University

2016 - 2017

Math 4181H: Honors Analysis I

Fall 2016

Math 4182H: Honors Analysis II

Spring 2017

RELEVANT COURSEWORK AND SKILLS

GMU Coursework: Optimization for machine learning, deep learning, computer vision, theory of computation, algorithms, graphics, software testing.

OSU Math Coursework: Real analysis, linear algebra, differential equations, probability, statistics, combinatorics, complex analysis, number theory, abstract algebra.

OSU CS Coursework: Software, digital logic, databases, operating systems, networking, theory of computation, machine learning, neural networks, natural language processing, GPU programming (CUDA).

Programming: Python (PyTorch, TensorFlow), Bash, Java, C, Git, Latex